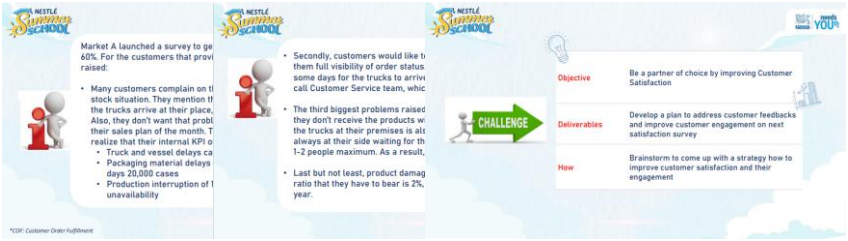
CASE STUDY



NESTLÉ SUMMER SCHOOL 2023

SUPPLY CHAIN FUNCTION – FINAL TEST

Market A launched a survey to get Customer Feedback. The response rate is only 60%. For the customers that provide the feedback, there are a lot of problems raised:

1/ Many customers complain on the lack of communication on Out Of Stock/Short stock situation. They mention that they only realize the short of the stock when the trucks arrive at their place, without any prior notice as well as alignment. Also, they don't want that problem happening too often like that as it will impact their sales plan of the month. The team has discussed about this feedback and realize that their internal KPI of COF is only **75%.**

- Truck and vessel delays caused late delivery of 50,000 cases to stocks
- Packaging material delays caused delays in planned production for 3 days 20,000 cases
- Production interruption of 15,000 cases due to raw and pack material unavailability

COF: Customer Order Fulfillment

2/ Secondly, customers would like to know when and where is their products, giving them full visibility of order status. They normally place an order, then wait for some days for the trucks to arrive. The only way for them to push for a date is to call Customer Service team, which is not convenient to them.

3/ The third biggest problems raised is Late Deliveries. 50% of the answers said that they don't receive the products within their expected lead-time. The offloading of the trucks at their premises is also being raised. They have to arrange loaders to always at their side waiting for the trucks, therefore, they can only affort to have 1-2 people maximum. As a result, it takes them more than 2 hours to fully offload.

4/ Last but not least, product damage is also one of their concerns. The bad goods ratio that they have to bear is 2%, which is abnormal compared versus 0.2% of last year.

EXPLAIN CASE

Objective	Be a partner of choice by improving Customer Satisfaction
Deliverables	Develop a plan to address customer feedbacks and improve customer engagement on next satisfaction survey
How	Brainstorm to come up with a strategy how to improve customer satisfaction and their engagement

Requirements

The main purpose of this case that how to **improve %Customer Satisfaction.**

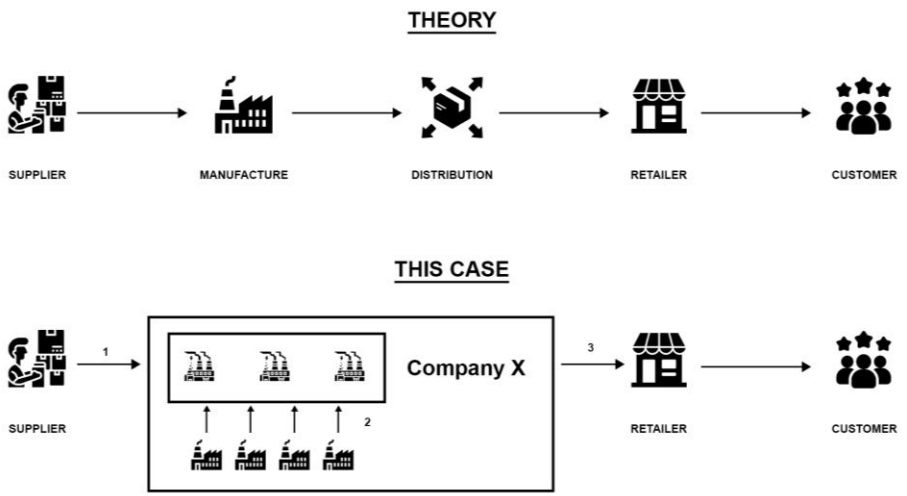
Structure and Prioritize Problem

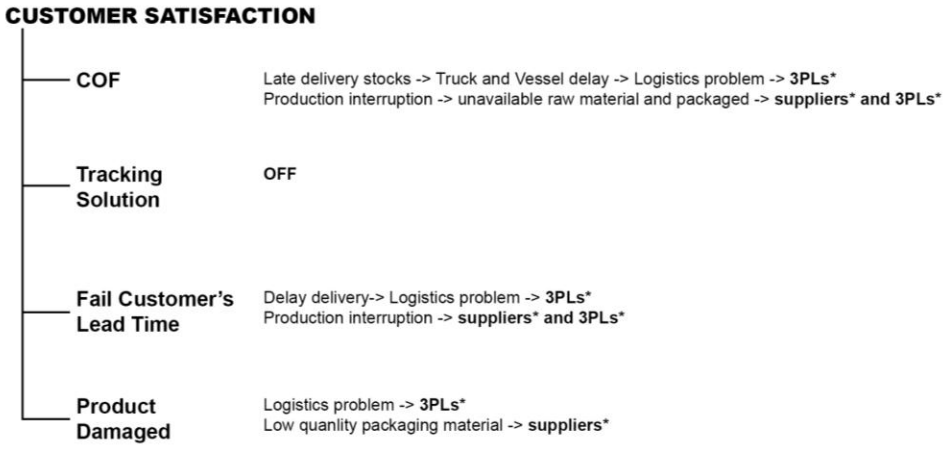
According to case study, there are four criteria with ranking category from more to less importance. Current information:

Customer Satisfaction: 60%, **COF**: 75%, **Tracking Solution**: NO

Fail Customer's Lead time: 50%, **Product Damaged**: 2%

EXPLAIN CASE





Suppose Supply Chain Flow

1. 100 Suppliers to Company X in two type material
2. 4 other Manufactures to Company X in stocks (final products)
3. Company X to Retailers

All of shipments were carried by 9 3PLs, Company X paid for them

Roots of issue

The roots of issue that the quality of supplier products and 3PLs services.

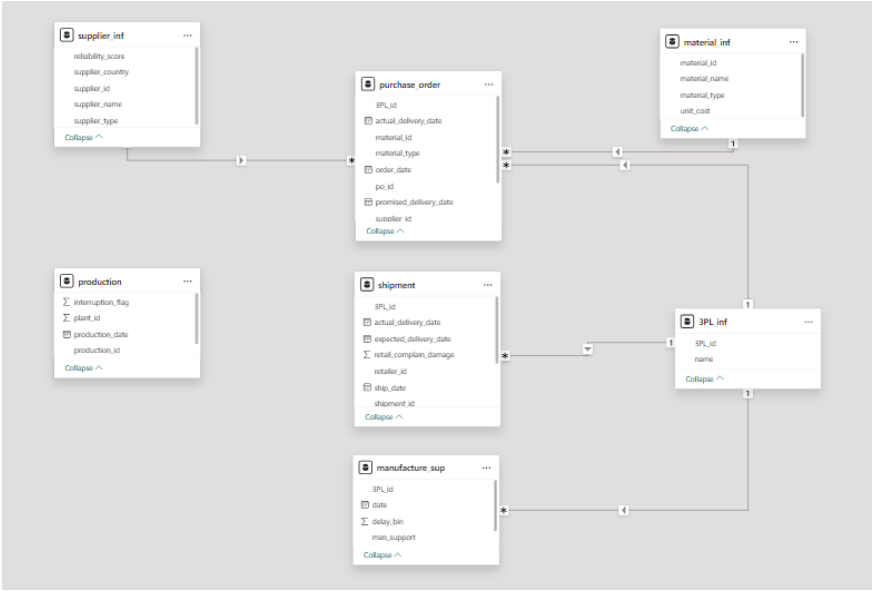
Suppose we have good suppliers, the main purpose of this case that **3PLs evaluation**.

DESCRIPTIVE

BOSS:

Customer Service Team provided the result of customer's survey that our **%Customer Satisfaction is only 60%**. What's happened? Let's check about COF, and OTIF.

Dataset name	Meaning	Basic information
supplier_inf	The information of supplier	100 suppliers information
material_inf	The information of materials	19 materials, category two type include raw material and packaged
3PL_inf	The information of Third Party Logistics	9 3PLs information
manufacture_sp	The stock flow from other manufacture to company X	90.000 records stock's shipments
purchase_order	The purchasing orders from supplier to company X	60.000 records purchasing orders
production	The production in company X	34.999 productions of 3 plants of company X
shipment	The information of shipments from company X to retailers	120.000 shipments to retailers
tracking	The status of tracking solution in company X	1 status of tracking solution, that's OFF



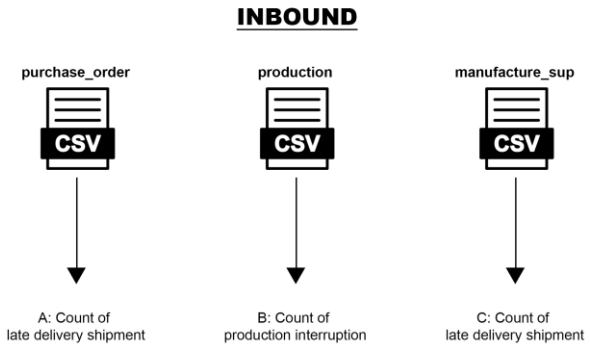
Created 2025 Simulation Dataset

To give a clear research method, I decided to create 8 datasets to records the case study clues

Make Relationship

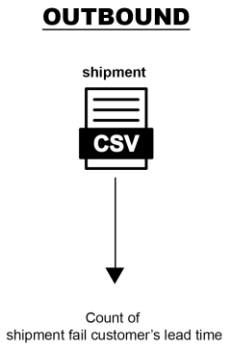
According to basic information of dataset, there are relationship of 8 datasets that were set up on Power BI.

DESCRIPTIVE

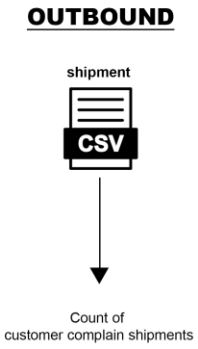


COF (%/yearly) = 1.308 * x^(-0.049)

Where: x = A + B + C
The explanation of function above in Predictive section



FAIL_CUS_LT (%) = C.of shipments FCLT / Total shipments



PRODUCT DAMAGE (%) = C.of customer complain shipments / Total shipments

Using DAX to create measures

To view the Customer Satisfaction's criteria in 2025, I created measures on Power BI.

DESCRIPTIVE

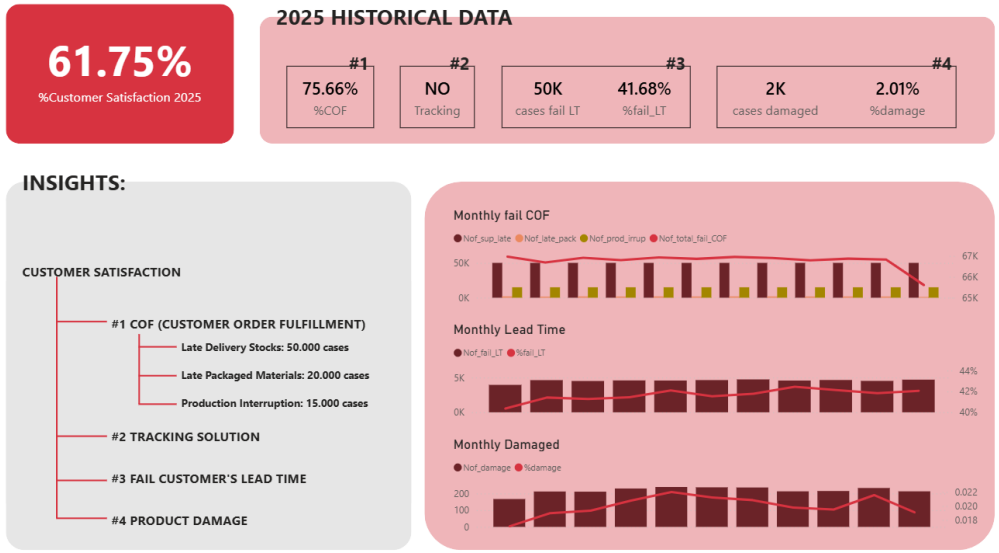
% CUSTOMER SATISFACTION = 0.62*COF + 0.18*tracking_solution + 0.12*on_time + 0.08*in_full

name	measurement	meaning	ranking	%share
COF	%		1	62%
tracking_solution	binary	on=1, off=0	2	18%
on_time	%	1 - %fail customer lead time	3	12%
in_full	%	1- %product damaged	4	8%

Customer Satisfaction Function

According to information in this case, I created the function base on the percentage sharing of criteria.

DESCRIPTIVE



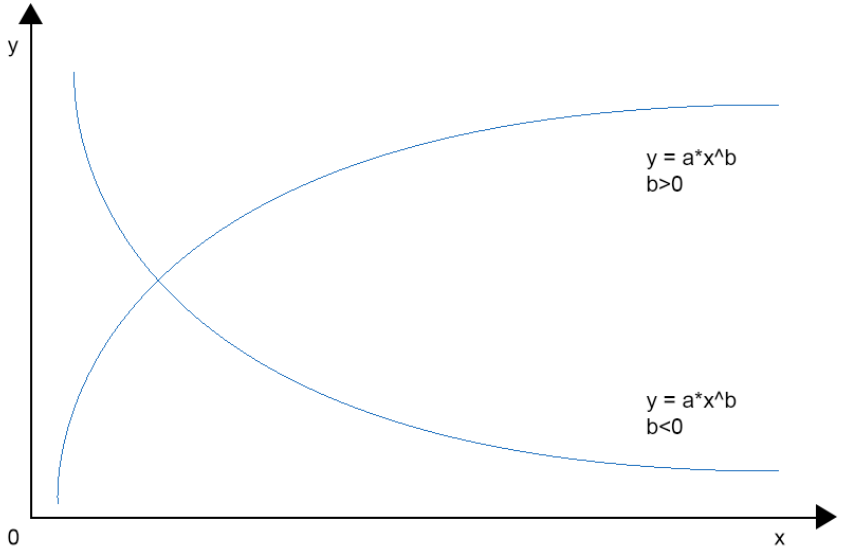
BI Descriptive Dashboard

Dashboard above shows **the summary of customer satisfaction performance** in 2025 (that nearly fit with the information of Nestlé case). Based on that, we can view the current **%Customer Satisfaction in only 61.75%.**

The other way to use descriptive dashboard that **apply new dataset** with same structure to view the details information month-by-month for tracking when apply new policy, method.

PREDICTIVE

BOSS:
Alright. We need to boost up our %Customer Satisfaction, and I want that is **as high as posible in the last 2026**. I'm only allow for 240 cases fail in COF, 120 cases fail in customer lead time, 24 cases fail in product damaged yearly. On the other hand, we will have **ERP solution in August 2026**, let's notice to customers.



Power Model

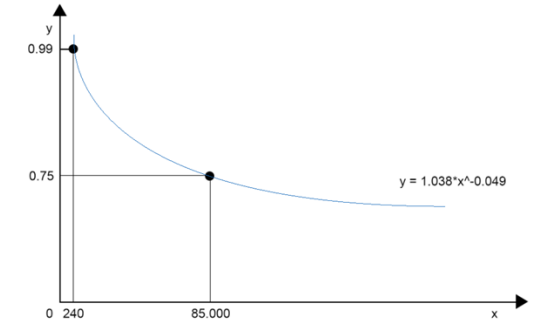
Because criteria are rating in percentage, to hit 100% is impossible. Therefore, they are following the **power function**.

PREDICTIVE

yi = ai*xi^bi
where i is criteria

xi	yi	yi meaning
Total fail of COF (3 measures in Descriptive section of inbound)	%COF	Customer Order Fulfillment
Count of shipments fail customer lead time	%on_time	Shipment meet customer lead time
Count of customer complain shipments	%in_full	Shipment isn't complain by customer because product damaged

%COF				
a	b	date	case_fail	COF
1.308	-0.049	2025	85000	75.00%
		2026	240	99.99%
%on_time				
a	b	date	case_fail	on_time
1.733	-0.115	2025	50000	49.94%
		2026	120	99.93%
%in_full				
a	b	date	case_fail	in_full
1.0135	-0.0043	2025	2400	98.01%
		2026	24	99.97%



Yearly Predictive Model

Based on the requirement, we have two information to find model that are historical data and target (calculate monthly to yearly by multiple with 12).

After that, we easily find three yearly models for criteria.

PREDICTIVE

yearly	y = a*x^b
monthly	y = a*12^b * x^b
COF example	
yearly	y = 1.308 * x^(-0.049)
monthly	y= 1.308 * 12^(-0.049) * x^(-0.049)

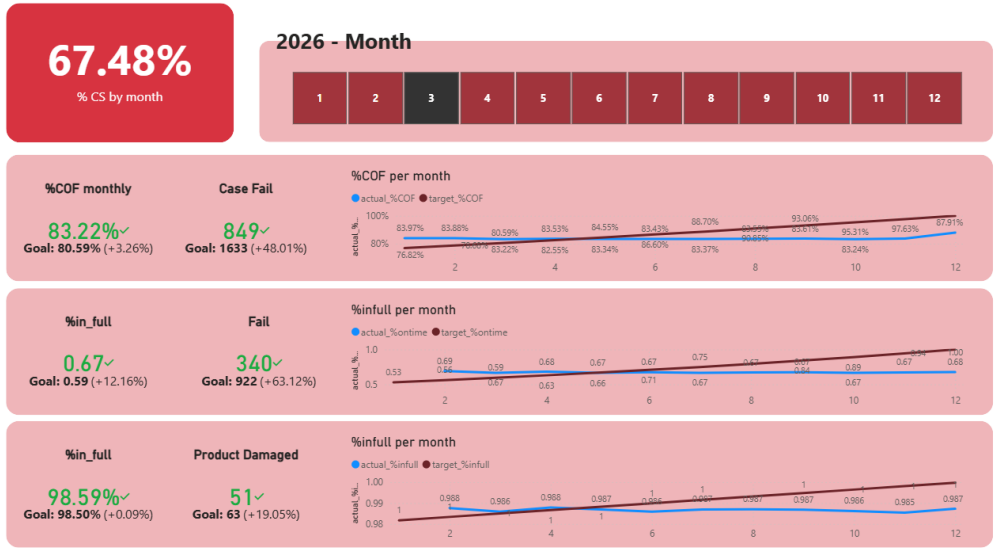
85.000 cases in 2025 => 7083 cases per month	7083*r^12=20 => r=0.61315, where 12 is number of month to hit target	
Dec-24	7083	75.00%
Jan-25	4343	76.82%
Feb-25	2663	78.68%
Mar-25	1633	80.59%
Apr-25	1001	82.55%
May-25	614	84.55%
Jun-25	376	86.60%
Jul-25	231	88.70%
Aug-25	142	90.85%
Sep-25	87	93.06%
Oct-25	53	95.31%
Nov-25	33	97.63%
Dec-25	20	99.99%

Predictive Model (Yearly to Monthly)

To set target monthly KPI, I transform yearly to monthly model. Then, I find the gap month-by-month by find **r value**. After that, I use the number of cases last month multiple with r value to get target this month.

In 2026, if the number of fail COF cases lower than target, it meet KPI.

PREDICTIVE

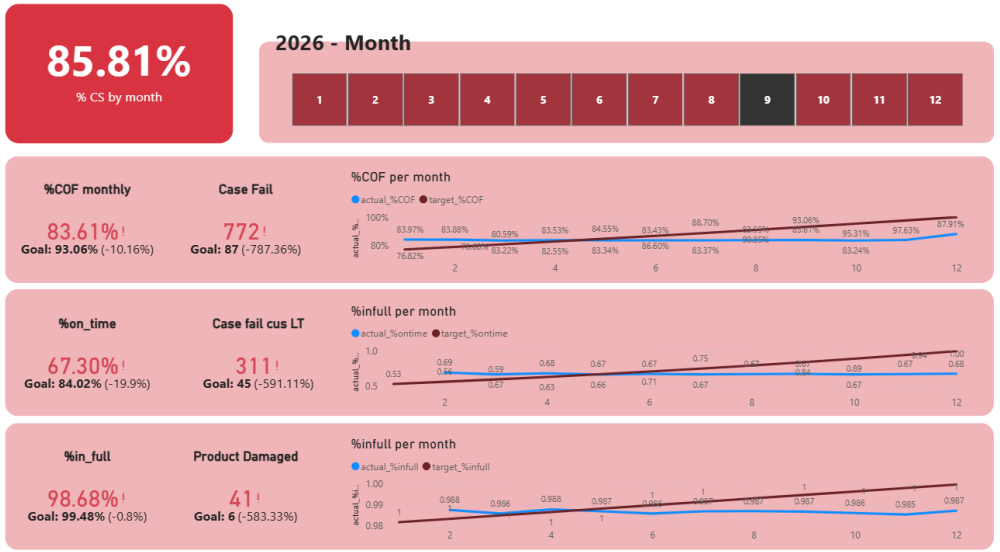


BI Dashboard

This dashboard shows the target of 3 criteria, such as %COF, %ontime, %infull and the %Customer Satisfaction, all of them present in monthly.

Besides, I created simulated 2026 dataset for target information in **KPI charts** and **Line charts** (for overall view). In fact, this dashboard can **automation upgrade** when I append the newest data to dataset. So, my boss can track 2026 performance day-by-day.

PREDICTIVE



Example: View in FEB-2026

In %KPI, three quarter meet target because the number of cases fail are lower than maximum cases fail.

But, in February 2026, company X don't have tracking solution, the Customer Satisfaction is slightly increase to 68.21%

Example: View in AUG-2026

Three criteria below don't meet target because the number of cases fail are out of limit. So, all color of them is red.

On the other hand, in August 2026, company X introduce the Tracking Solution for their customers, so Customer Satisfaction improve to 85.81%.

PRESCRIPTIVE

BOSS:

The main problem was caused by 3PLs. You should to provide the plan to distribute the number of shipments for different 3PLs that based on their performance to hit the high %Customer Satisfaction.

3PL INFORMATION	INTERNAL SHIPMENT IN MARKET A			GLOBAL SHIPMENT TO MARKET A	
3PL_id	total_trucking_shipmer	trucking_late	product_damage	total_global_shipment	global_late
3pl_001	13446	5589	275	16651	4738
3pl_002	13391	577	202	17009	8804
3pl_003	13206	9045	344	16475	6528
3pl_004	13389	1085	184	16532	10534
3pl_005	13422	5565	302	16709	3710
3pl_006	13423	10098	361	16695	11791
3pl_007	13261	11088	292	16570	5682
3pl_008	13207	5485	254	16463	11486
3pl_009	13255	1486	192	16896	6815
	120000	50018	2406	150000	70088

3PL_id	%trucking_ontime	%trucking_infull	%global_ontime
3pl_001	58.43%	97.95%	71.55%
3pl_002	95.69%	98.49%	48.24%
3pl_003	31.51%	97.40%	48.24%
3pl_004	91.90%	98.63%	36.28%
3pl_005	58.54%	97.75%	77.80%
3pl_006	24.77%	97.31%	29.37%
3pl_007	16.39%	97.80%	65.71%
3pl_008	58.47%	98.08%	30.23%
3pl_009	88.79%	98.55%	59.67%

Overall 2025 historical data

Using Power Query to combine datasets to a table above.

Trucking: that is regional transport in Market A (company X to retailers/ customers). Product flow number 3 in Descriptive section.

Global shipment: that is international transport to Market A (suppliers/ manufactures to Company X). Product flow number 1, 2 in Descriptive section

Explain to performance

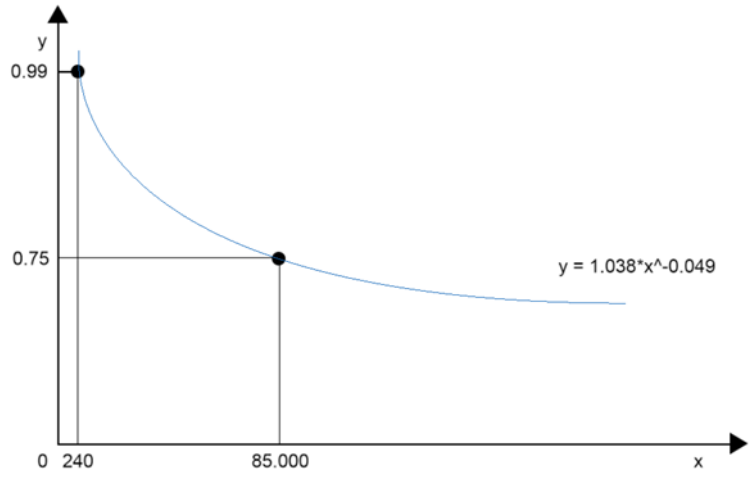
Using performance function that: **1 – (a/b)**

Where: i is the total of shipments

j is the cases fail (late delivery, damage)

So, we can rank 3PLs based on their performance in 2025.

PRESCRIPTIVE



Customer Satisfaction	**60.33%**

	cases fail	%
%COF	85000	75.00%
Tracking Solution	NO	0.00%
%on_time	50018	49.93%
%in_full	2406	98.01%

Set Objective Function

As we know that my boss want to improve %Customer Satisfaction. Clearly, %CS depended four criteria below. If the number of cases fail is decrease, %criteria will improve according power model then %CS will also increase.

So that, the objective function in this solver model **that minimum cases fail of each category**.

PRESCRIPTIVE

3PL INFORMATION	INTERNAL SHIPMENT IN MARKET A			GLOBAL SHIPMENT TO MARKET A	
3PL_id	**total_trucking_shipment**	**trucking_late**	**product_damage**	**total_global_shipment**	**global_late**
3pl_001		0	0		0
3pl_002		0	0		0
3pl_003		0	0		0
3pl_004		0	0		0
3pl_005		0	0		0
3pl_006		0	0		0
3pl_007		0	0		0
3pl_008		0	0		0
3pl_009		0	0		0
Total	**120000**	**0**	**0**	**150000**	**0**
Change with non-optimize					
% Change with non-optimize					

Set Variables

The orange cells is variables, that is the number of shipments that company X separated to nine 3PLs. Based on 3PLs performance, we can calculate the number of cases fail like late delivery and damage.

PRESCRIPTIVE

1. Separate the volume of shipments
In terms of on time criteria, according Pareto Theory, 80% volume was handled by top 3 3PLs, 20% volume was handled by others

Total top 3 share volume	Optimal	Must greater than	Must less than
Trucking shipments	0.00%	77%	83%
Global shipments	0.00%	77%	83%

2. Keep MOQ shipments to get lower logistics cost, and the number of 3PLs
To avoid break product flow in supply chain, company X decided to keep all of them.
3PLs provide different productivities per month with different logistics cost.
Each 3PLs want 30 shipments per month at least to provide promotional logistics cost. (30 shipments/ month = 360 shipments/ year)

$xi1 + xi2 > 360$ where i is 3PLs
1 is trucking shipment, 2 is global shipment

3PL_id	total_trucking_shipment	total_global_shipment	total_shipment	Must greater than
3pl_001	0	0	0	360
3pl_002	0	0	0	360
3pl_003	0	0	0	360
3pl_004	0	0	0	360
3pl_005	0	0	0	360
3pl_006	0	0	0	360
3pl_007	0	0	0	360
3pl_008	0	0	0	360
3pl_009	0	0	0	360

4. Fit with 2025 data
Total trucking and global shipments must equal to 2025 data

	Optimal	2025
Trucking shipment	0	120,000
Global shipment	0	150,000

5. Mathematics
$xi1, xi2 > 0$ and is integer

Set Constraints

Based on boss's requirements, there are four important constraints to keep product flow fluency.

PRESCRIPTIVE

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3pl_009	13255	1486	192	16896	6815
	120000	50018	2406	150000	70088

3PL INFORMATION	INTERNAL SHIPMENT IN MARKET A			GLOBAL SHIPMENT TO MARKET A	
3PL_id	total_trucking_shipment	trucking_late	product_damage	total_global_shipment	global_late
3pl_001	360	150	7	0	0
3pl_002	99240	4276	1497	0	0
3pl_003	0	0	0	360	186
3pl_004	360	29	5	0	0
3pl_005	0	0	0	124140	27564
3pl_006	0	0	0	360	254
3pl_007	0	0	0	360	123
3pl_008	20040	8323	385	0	0
3pl_009	0	0	0	24780	9995
Total	**120000**	**12778**	**1895**	**150000**	**38123**
Change with non-optimize		**-37240**	**-511**		**-31965**
% Change with non-optimize		**-74.45%**	**-21.25%**		**-45.61%**

Result

As a result, the output of objective function is **52,795 cases fail**

Especially, we can see the improvements of each different cases fail. It is boosted up!

PRESCRIPTIVE

Customer Satisfaction		**60.33%**
	cases fail	%
%COF	85000	75.00%
Tracking Solution	NO	0.00%
%on_time	50018	49.93%
%in_full	2406	98.01%

			Change with non-optimize
Customer Satisfaction		**67.02%**	+6.68%
	cases fail	%	
%COF	8169	84.12%	+9.12%
Tracking Solution	NO	0.00%	0.00%
%on_time	12778	58.42%	+8.48%
%in_full	1895	98.11%	+0.10%

Conclusion

As conclusion, we can see that there are improvements in %Customer Satisfaction and criteria when we distributed different the number of shipments to 3PLs.

Some Advise

In this case, to maximum the %Customer Satisfaction, we need to manage more than 9 3PLs, reach to good 3PLs to meet customer demand on time in full and our production plan.

Out of that case, we need to consider other criteria like supplier's quality, plant production,… to improve %Customer Satisfaction continuously.